BASIC MECHANICAL ENGINEERING

Subject Code: 23A01201T (Part B) | UNIT – II : MANUFACTURING PROCESSES & THERMAL ENGINEERING

1. Manufacturing – Introduction

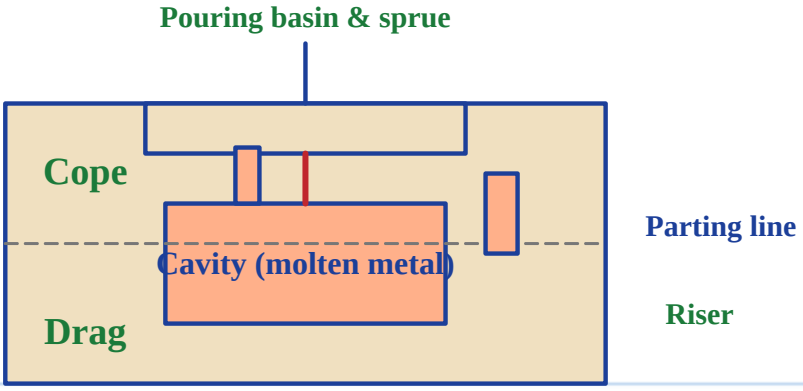
Manufacturing = converting **raw materials** into **finished useful products** through a sequence of processes (using machines, tools, labour, energy).

Manufacturing processes → Casting / Forming / Joining / Machining / Additive (3D printing)

- Selection depends on material, shape, size, quantity, accuracy, surface finish, cost.

2. Casting

Casting: pouring **molten metal** into a **mould cavity** of the required shape and allowing it to **solidify**; the solid piece is the **casting**.



Sand mould (two-part box)

Fig 1.1 Sand casting mould

❖Principle & steps

- **Pattern making** (wood/ metal/ wax/ plastic, with allowances: shrinkage, machining, draft, shake, distortion) → **mould making** (sand mould with gating system: **pouring basin**, **sprue**, **runner**, **gate**; **riser** to feed shrinkage; **cores** make hollow parts) → **melting** (cupola, induction/ electric arc furnace) → **pouring** → **solidification & cooling** → **shake-out/ fettling** (cleaning) → **inspection**.
- **Pattern** = replica of the part (made slightly larger); **mould** = cavity; **core** = sand shape forming internal cavities; **cope** (top box), **drag** (bottom box), **parting line**.

- **Mould sand:** silica sand + clay (bentonite) + water; needs permeability, strength, refractoriness.

❖Types of casting

- **Sand casting** – cheap, any size/ metal (engine blocks, pipes). **Die casting** (permanent metal die; pressure die – Al, Zn parts: carburettors, gear housings) – high precision, fast. **Investment (lost wax) casting** – intricate, high finish (turbine blades, jewellery). **Centrifugal casting** – pipes, liners. **Shell moulding, continuous casting** (steel billets).

❖Advantages / limitations

- Complex shapes, any size (few grams to tonnes), cheap for mass production, wide material choice. **Defects:** blow-holes, shrinkage cavities, cold shuts, misruns, hot tears, porosity; needs finishing; lower strength than forged.
- **Applications:** engine blocks, cylinder heads, pistons, pump casings, machine beds, bells, statues, pipes, wheels, valves.

3. Forming (Metal Working)

> **Forming** = **plastic deformation** of metal into the required shape by applying force, **without removing material** (no chips); volume remains constant.

- **Hot working** (above recrystallisation temperature – easy shaping, no strain hardening) vs **Cold working** (below – better strength & finish, closer tolerance, needs more force).

❖Main forming processes

- **Forging:** compressing heated metal with **hammers/ presses/ dies** – improves grain flow & strength (crankshafts, connecting rods, spanners, axes). Types: open die, closed (impression) die, drop, press, upset.
- **Rolling:** passing metal between **rotating rolls** to reduce thickness/ make shapes – plates, sheets, rails, I-beams, bars (hot & cold rolling).
- **Extrusion:** forcing metal through a **die** to get long constant section (pipes, rods, Al window frames, tubes). Direct/ indirect.
- **Drawing:** pulling metal through a die to reduce diameter – **wire drawing**, tube drawing, **deep drawing** (cups, utensils, car panels).
- **Sheet-metal operations:** shearing, blanking, punching, bending, stamping/ pressing, spinning.

- Other: powder metallurgy (compacting & sintering metal powders – gears, bearings, tungsten filaments), rolling of threads.

❖Comparison

Casting	Forming
Liquid metal	Solid metal deformed
Complex shapes	Simpler shapes
Lower strength	Higher strength (grain flow)
More defects	Good integrity

4. Joining Processes

> Joining = connecting two or more parts permanently or temporarily.

- Temporary (mechanical): bolts, nuts, screws, rivets (semi-permanent), keys, pins, cotters, couplings.
- Permanent: welding, brazing, soldering, adhesive bonding, riveting.

❖Welding

- Welding = joining metals by heat (and/ or pressure), usually with a filler metal, forming a fusion between them.
- Fusion welding (melt base metal): Arc welding (electric arc 3000–4000 °C; SMAW/ stick, MIG/ GMAW, TIG/ GTAW, submerged arc) – gas welding (oxy-acetylene flame ≈ 3200 °C) – thermit welding (railways), laser, electron beam.
- Solid-state / pressure welding: resistance welding (spot, seam) – body panels; friction (stir) welding, forge welding, ultrasonic.
- Brazing (filler > 450 °C, base metal does not melt – brass filler for Cu/ steel), Soldering (filler < 450 °C – Pb-Sn, electronics, pipes).
- Arc welding equipment: power source (transformer/ rectifier/ inverter), electrode & holder, earth clamp, cables, helmet (shade), gloves (safety!).
- Weld defects: porosity, slag inclusion, cracks, undercut, incomplete penetration, distortion.
- Applications: ships, pressure vessels, pipelines, bridges, car bodies, structures, machinery repair.
- Advantages: strong, leak-proof joints, light structure, versatile. Disadvantages: residual stress, needs skill, permanent, HAZ (heat-affected zone).

5. Machining

> **Machining** = removal of excess material in the form of chips from a workpiece using a cutting tool on a machine tool to obtain accurate shape, size and surface finish.

❖Cutting parameters

- Cutting speed (V) = π D N / 1000 m/min; Feed (f) – tool advance per revolution/ stroke (mm/rev); Depth of cut (d) (mm). Material removal rate MRR = V f d.

❖Basic machine tools & operations

Machine	Motion (work / tool)	Operations
Lathe	Work rotates, tool moves linear	Turning, facing, drilling, boring, threading, knurling, parting
Drilling machine	Drill rotates & feeds	Drilling, reaming, tapping, counter-sinking
Shaper / Planer	Tool reciprocates (shaper)	Flat surfaces, slots, keyways
Milling machine	Multi-tooth cutter rotates; work feeds	Slots, gears, flat/ contoured surfaces
Grinding machine	Abrasive wheel	Fine finish, tool sharpening
Broaching, sawing, boring		Keyways, cut-off

- Cutting tool materials: HSS, carbides (tungsten carbide), ceramics, CBN, diamond; cutting fluids (coolant & lubricant).
- Tool geometry: rake angle, clearance (relief) angle, nose radius.
- Non-conventional machining: EDM (spark erosion), ECM, USM, laser, abrasive water jet – for hard materials/ complex shapes.
- Finishing: polishing, honing, lapping, buffing.

6. CNC Machines

> **CNC (Computer Numerical Control)** = machine tools whose movements are controlled automatically by a computer executing a pre-programmed set of coded instructions (G-code & M-code).

- **Parts:** Part program (G-code), MCU (machine control unit/ computer), drives & servomotors, feedback (encoders, resolvers), machine tool structure (slides, spindle, ball screws), tool changer (ATC), coolant.
- **Axes:** X, Y, Z (+ A, B, C rotary); 3-axis, 5-axis machining centres.
- **Types:** CNC lathe (turning centre), CNC milling/ VMC & HMC, CNC drilling, grinding, EDM, laser/ plasma cutting, routers, CNC press brake.
- **Open loop vs closed loop control:** closed loop uses feedback → accuracy.
- **G-codes:** G00 rapid move; G01 linear feed; G02/ G03 circular arc CW/ CCW; G90 absolute; G91 incremental. **M-codes:** M03 spindle ON clockwise; M05 stop; M06 tool change; M30 program end.
- **Advantages:** high accuracy & repeatability, complex shapes, less manual skill, reduces setup time, consistent quality, 24 × 7 production, safety.
- **Disadvantages:** high initial & maintenance cost, skilled programmer needed, power dependent.
- **Applications:** automotive & aerospace parts, moulds & dies, medical implants, electronics.

7. 3D Printing (Additive Manufacturing)

> Additive manufacturing (3D printing) builds a part layer by layer from a 3D CAD model by adding material, instead of removing it (subtractive).

❖Process steps

- CAD model → STL file → slicing (layers) → printing → post-processing (support removal, curing, polishing).

❖Technologies

- FDM/ FFF (fused deposition – melted plastic filament: PLA, ABS); SLA (stereolithography – UV laser cures liquid resin); SLS (laser sinters powder – nylon); SLM/ DMLS (metal powder bed – Ti, steel, Inconel); Binder jetting; LOM; Electron beam melting; WAAM; concrete 3D printing (houses).

❖Advantages

- Complex shapes, rapid prototyping, customisation, low waste, no tooling, light-weight lattice structures, on-demand spares.

❖Limitations

- Slow for mass production, limited size/ materials, surface finish & strength anisotropy, high cost for metals.

❖Applications

- Prototyping, aerospace (fuel nozzles), medical (implants, dental, prosthetics), automotive, jewellery, tooling, construction, education.

8. Smart Manufacturing (Industry 4.0)

> Smart manufacturing = using connected digital technologies so that factories sense, analyse, decide and adapt automatically.

- Key technologies: IoT sensors, Big data & AI/ ML, cloud/ edge computing, cyber-physical systems, digital twin (virtual copy of machine/ plant), robots & cobots, AR/ VR, 5G, additive manufacturing, MES/ ERP integration, predictive maintenance.
- Benefits: higher productivity, quality & flexibility, less downtime & waste, energy saving, mass customisation, safety, real-time monitoring.
- Challenges: investment, cyber-security, skilled workforce, interoperability.
- Industrial revolutions: 1.0 steam/ mechanisation → 2.0 electricity/ mass production → 3.0 electronics/ automation (PLC, CNC) → 4.0 cyber-physical/ smart.

9. THERMAL ENGINEERING – Basics

- Thermodynamics: science of energy – heat, work. System, surroundings, closed/ open systems.
- First law: $Q = \Delta U + W$ (energy is conserved). Second law: heat cannot flow from cold to hot by itself; no engine is 100% efficient (Kelvin–Planck); Clausius statement for refrigerators.
- Heat engine: receives heat Q_1 from source, converts part into work W, rejects Q_2 to sink: $\eta = W/Q_1 = 1 - Q_2/Q_1$; Carnot $\eta = 1 - T_2/T_1$.
- Refrigerator: takes heat from cold space using work; $COP = Q_L/W$.

10. Boilers

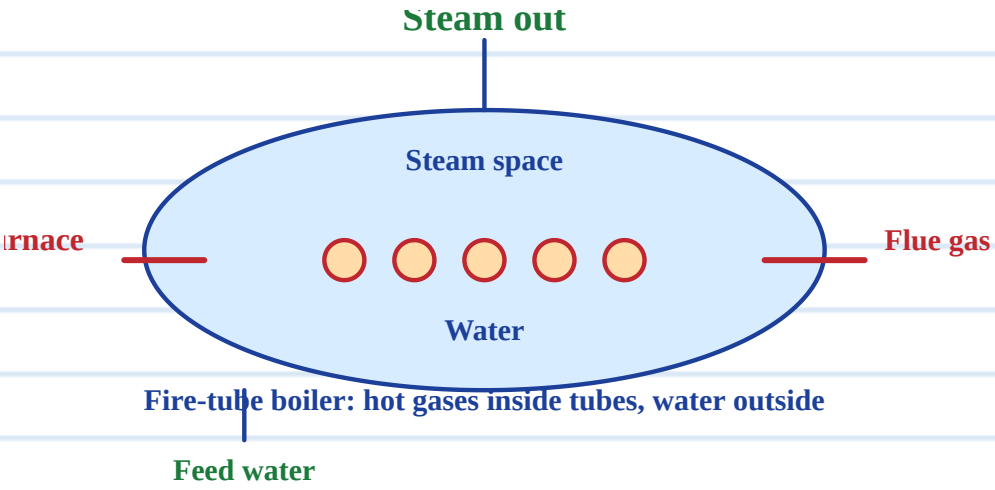


Fig 2.3 Fire-tube boiler (Cochran/ Lancashire principle)

> A boiler (steam generator) is a closed vessel in which water is heated to produce steam at the required pressure and temperature, using heat from fuel combustion.

❖Classification

- Fire-tube boilers: hot gases flow through tubes surrounded by water – simple, low pressure (≤ 20 bar), low capacity; e.g., Cochran, Lancashire, locomotive, Scotch marine.
- Water-tube boilers: water flows in tubes, hot gases outside – high pressure (up to 250 bar), large capacity, quick steam; e.g., Babcock & Wilcox, Stirling; used in power plants.
- Others: vertical/ horizontal, internally/ externally fired, natural/ forced circulation (Benson, La Mont), supercritical.

❖Working principle (fire-tube)

- Fuel burns in furnace → hot flue gases pass through the tubes → heat transferred to surrounding water → water boils → steam collects in steam space → leaves through steam stop valve to the engine/ turbine; feed water pumped in; flue gas exits via chimney.

❖Boiler mountings (safety & control – on the boiler)

- Water level indicator, pressure gauge, safety valve, steam stop valve, feed check valve, blow-off cock, fusible plug, man-hole.

❖Boiler accessories (improve efficiency)

- Economiser (heats feed water using flue gas), air pre-heater, superheater (raises steam temperature above saturation), feed pump, injector.

❖Fire-tube vs water-tube

Feature	Fire-tube	Water-tube
Hot gas	Inside tubes	Outside tubes
Pressure	Low (≤ 20 bar)	High (≤ 250 bar)
Capacity	Small	Large
Steam raising	Slow	Quick
Risk	Heavy explosion	Less (small tubes)
Cost	Low	High

- Boiler efficiency η = (steam generated × $(h_s - h_f)$) / (fuel × CV) ≈ 75–90%. Equivalent evaporation.

II. Internal Combustion (IC) Engines

> IC engine = heat engine in which fuel is burned inside the engine cylinder, and the high-pressure gases push a piston, which turns the crankshaft through a connecting rod.

❖Main components

- Cylinder block, cylinder head, piston, piston rings, gudgeon pin, connecting rod, crankshaft, flywheel, camshaft, inlet & exhaust valves, spark plug (SI) / fuel injector (CI), carburettor/ fuel pump, manifold, crankcase, gaskets, cooling & lubrication systems.

❖Terms

- TDC & BDC (top & bottom dead centre), stroke (L), bore (D), swept (displacement) volume $V_s = (\pi/4)\, D^2 L$, clearance volume V_c, compression ratio $r = (V_s + V_c)/V_c$ (SI 6–10; CI 15–22).

❖Classification

- By fuel: petrol, diesel, gas (CNG/ LPG), bio-fuel; by cycle: Otto, Diesel; by strokes: 2-stroke, 4-stroke; by ignition: SI/ CI; by cylinders: single/ multi; cooling: air/ water; arrangement: inline, V, radial.

12. Four-Stroke Engine

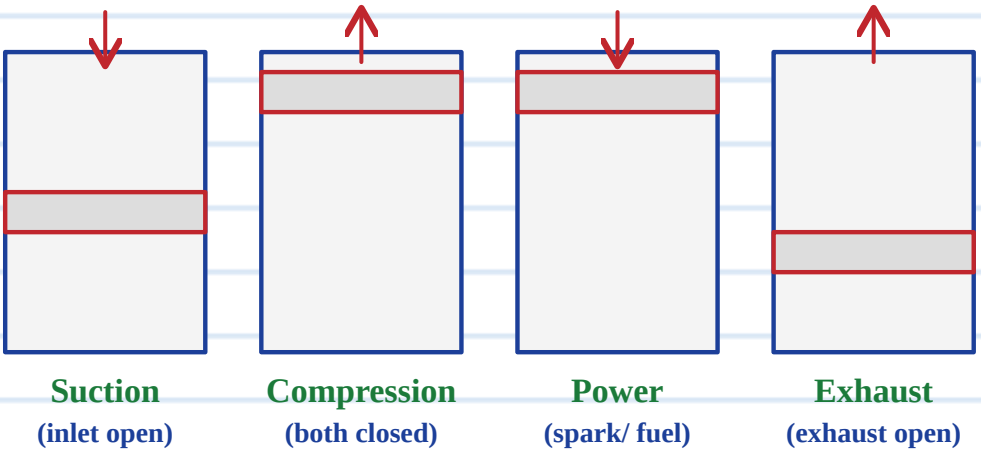


Fig 2.1 Four strokes: piston down-up-down-up (2 revolutions per cycle)

- **Cycle completed in 4 piston strokes (2 crank revolutions = 720°);** one power stroke per 2 revolutions.
- **1. Suction (intake):** piston TDC → BDC; **inlet valve open;** air-fuel mixture (petrol) or only air (diesel) drawn in.
- **2. Compression:** BDC → TDC; both valves closed; charge compressed (petrol: spark at end; diesel: air heated to 500–600 °C).
- **3. Power (expansion):** burning gases push piston TDC → BDC; **only power-producing stroke;** both valves closed.
- **4. Exhaust:** BDC → TDC; **exhaust valve open;** burnt gases expelled.
- Valves operated by **camshaft** (rotates at half crank speed).
- Applications: cars, buses, trucks, tractors, generators.

13. Two-Stroke Engine

- **Cycle completes in 2 strokes (1 revolution = 360°);** one power stroke every revolution. **No valves – ports (inlet, transfer, exhaust)** uncovered by piston.
- **Upward stroke:** piston compresses charge above, simultaneously **fresh charge is drawn into crankcase** (through inlet port). Ignition near TDC.
- **Downward stroke (power):** expansion; piston uncovers exhaust port → burnt gas out; then **transfer port** opens, **crankcase charge pushed into cylinder** (scavenging).
- Applications: mopeds, scooters, chain-saws, outboard motors, small generators.

❖2-stroke vs 4-stroke

Feature	2-stroke	4-stroke
Power strokes	Every revolution	Every 2 revolutions
Valves	Ports	Valves + cam
Power for same size	More (1.5–1.8×)	Less
Fuel economy / thermal efficiency	Lower (charge loss)	Higher
Lubrication	Mixed oil in fuel	Separate sump
Pollution	More	Less
Weight, cost	Light, cheap	Heavier, costly
Smoothness	Less	More

14. SI & CI Engines

Feature	SI (Petrol – Otto)	CI (Diesel)
Fuel	Petrol (high volatility, high octane)	Diesel (high cetane no.)
Mixture formation	Carburettor/ MPFI (outside)	Injector (inside cylinder)
Ignition	Spark plug	Compression (self) ignition
Compression ratio	6 – 10	15 – 22
Cycle	Otto (constant volume heat addition)	Diesel (constant pressure)
Thermal efficiency	25 – 30%	35 – 40%
Weight/ cost	Lighter, cheaper	Heavier, costlier
Noise, vibration	Less	More (knock)
Pollutants	CO, HC	NO_x, PM (soot)
Use	Cars, bikes	Trucks, buses, tractors, ships

- Knocking – SI: auto-ignition of end gas (use high octane); CI: delay period (use high cetane).
- Octane rating (SI), Cetane number (CI); Compression ignition requires T > 250–300 °C self-ignition of diesel.

15. Air-Standard Cycles

❖Otto cycle (constant volume) – petrol engine

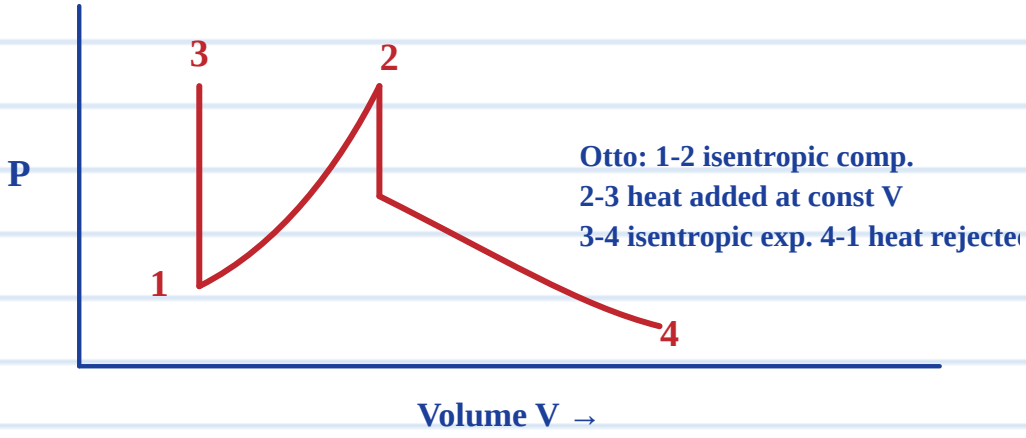


Fig 2.2 P-V sketch of an Otto cycle

- Processes: 1–2 isentropic compression; 2–3 constant volume heat addition (spark); 3–4 isentropic expansion (power); 4–1 constant volume heat rejection.

$$\eta_{Otto} = 1 - 1 / r^{\gamma-1}$$ (r = compression ratio, $\gamma = 1.4$ for air)

- Efficiency increases with r (limited by knocking). Example: $r = 8$: $\eta = 1 - 1/8^{0.4} = 1 - 0.435 = 56.5\%$ (ideal; actual 25–30%).

❖Diesel cycle (constant pressure) – diesel engine

- 1–2 isentropic compression; 2–3 constant pressure heat addition (fuel injection & burning); 3–4 isentropic expansion; 4–1 constant volume heat rejection.

$$\eta_{Diesel} = 1 - [\,1 / r^{\gamma-1}\,] \times [\,(\rho^{\gamma} - 1) / (\gamma\,(\rho - 1))\,]$$, ρ = cut-off ratio = V_3/V_2

- For the same compression ratio, $\eta_{Otto} > \eta_{Diesel}$; but Diesel works at higher r (15–22), so actual diesel engines are more efficient.
- Example: $r = 18$, $\rho = 2$: $\eta = 1 - (1/18^{0.4})[(2^{1.4}-1)/(1.4\times1)] = 1 - 0.315\times[1.639/1.4] = 1 - 0.369 = 63.1\%$.
- Dual cycle – part heat at constant V, part at constant P (modern high-speed diesel).
- Performance: indicated power $IP = p_m L A N n / 60$; brake power $BP = 2\pi NT/60$; mechanical efficiency = BP/IP; brake thermal efficiency = $BP/(\dot{m}_f \times CV)$.

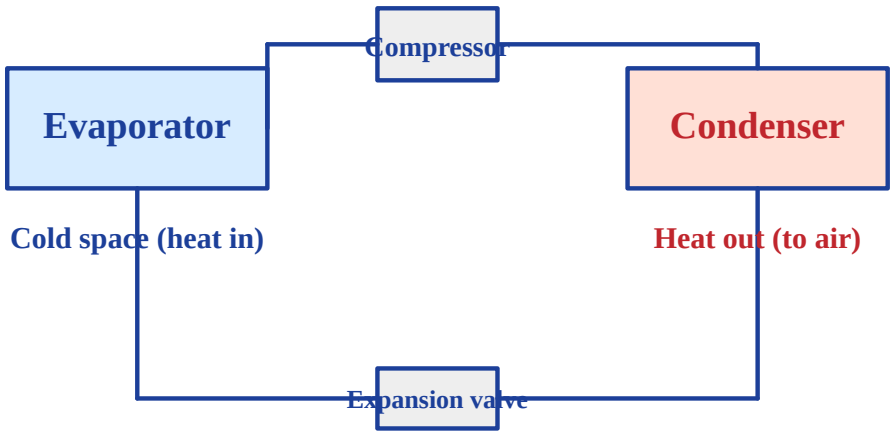
16. Refrigeration

Refrigeration = process of removing heat from a low-temperature space/ substance and rejecting it to a higher-temperature surrounding (needs external work). Device = **refrigerator** (cooling) / **heat pump** (heating).

- **Unit:** 1 ton of refrigeration (TR) = 3.5 kW = 211 kJ/min (heat to freeze 1 tonne of water at 0 °C into ice in 24 h).

COP = Refrigerating effect / Work input = Q_L / W ; Carnot COP = T_L / (T_H – T_L)

❖Vapour Compression Refrigeration (VCR) cycle



Vapour compression refrigeration cycle

Fig 2.4 Vapour compression refrigeration system

- **Compressor:** low-pressure vapour refrigerant → high-pressure, high-temperature vapour (work input).
- **Condenser:** vapour rejects heat to surrounding air/ water and **condenses to liquid.**
- **Expansion valve / capillary tube:** throttling – pressure & temperature drop (partial flash).
- **Evaporator:** cold liquid absorbs heat from the space and **evaporates** → cooling effect; vapour returns to compressor.
- **Refrigerants:** R-134a, R-600a (isobutane), R-290, ammonia (NH_3), CO_2; old CFC/ R-12 banned (**ozone depletion**, Montreal protocol); low GWP preferred.
- **Applications:** domestic refrigerator, cold storage, ice plants, food & medicine preservation, transport.

❖Vapour absorption cycle (brief)

- Uses heat (not work) with NH_3–water or LiBr–water: absorber, pump, generator, condenser, expansion valve, evaporator; suited where waste heat/ solar heat is available; lower COP (~0.6).

17. Air-Conditioning

> Air-conditioning (AC) = process of simultaneously controlling temperature, humidity, cleanliness and distribution of air in a space to meet comfort or process requirements.

- Comfort conditions: temp 22–26 °C, relative humidity 40–60%, air velocity ≈ 0.15–0.25 m/s.

❖Basic processes (psychrometry)

- Sensible cooling/ heating (change in dry bulb temperature); humidification/ dehumidification; cooling & dehumidification; heating & humidification; mixing.
- Psychrometric terms: dry-bulb temp (DBT), wet-bulb temp (WBT), dew point temp (DPT), relative humidity (RH), specific humidity (ω), enthalpy.

❖Working of a window/ split AC

- VCR cycle with the evaporator coil (indoor) cooling & dehumidifying room air blown by a fan; condenser (outdoor) rejects heat; filter cleans air; thermostat; inverter AC varies compressor speed.
- Ratings: tonnage (1.0, 1.5 TR), BEE star rating (ISEER), EER/ COP.

❖Types of AC systems

- Window, split, ductable, VRF/ VRV, chiller (central), packaged unit, evaporative (desert) cooler. Winter AC = heating + humidifying.
- Applications: homes, offices, malls, theatres, hospitals (OT), data centres, trains, vehicles, clean rooms, textile industries.

18. Components of Electric & Hybrid Vehicles

❖Electric Vehicle (EV / BEV)

> An EV uses an electric motor powered by a battery pack (no engine, zero tail-pipe emission).

- Battery pack (Li-ion, LFP/ NMC; 20–100 kWh) – stores energy; BMS (battery management system) monitors voltage, temperature, SOC, balancing.
- Electric motor (BLDC, PMSM, induction) – efficiency ~ 90%+; instant torque.
- Power electronics: inverter (DC → AC for motor), DC–DC converter (for 12 V auxiliaries), motor controller.
- On-board charger (OBC) + charging port (AC slow, DC fast – CCS/ CHAdeMO/ Bharat DC-001).
- Transmission: single-speed reduction gear/ differential (no multi-gear gearbox, no clutch).
- Regenerative braking: motor acts as generator in braking → recovers energy back to battery.
- Thermal management system (battery cooling), vehicle control unit (VCU), display/ telematics, HV cables & contactors, fuse.
- Advantages: zero emissions locally, high efficiency, quiet, low running cost, less maintenance. Limitations: range & charging time, battery cost/ weight, charging infrastructure, recycling.

❖Hybrid Electric Vehicle (HEV)

- Combines IC engine + electric motor + battery.
- Types: Series hybrid (engine only charges battery via generator; motor drives wheels – e.g., range extender); Parallel hybrid (both engine and motor drive wheels); Series–parallel / power-split (Toyota Prius); Mild hybrid (assist only), Full hybrid, Plug-in hybrid (PHEV – external charging).
- Extra components: engine, motor-generator, power-split device/ planetary gear, battery (small), controller, regenerative braking.
- Benefits: 25–40% better mileage, lower emissions, longer range than EV. Limits: complexity & cost.

❖EV vs IC vehicle

Feature	EV	IC engine vehicle
Energy source	Battery	Petrol/ diesel
Emission	Zero tail-pipe	CO_2, NO_x, PM
Efficiency	≈ 85–90%	25–40%
Moving parts	Few	Many
Refuel time	30 min – 8 h	3 min
Noise	Very low	High

- Other: Fuel-cell EV (FCEV) – $H_2 + O_2 \rightarrow$ electricity; India's FAME-II scheme, PLI, National EV mission.

19. Solved Problems

❖Problem 1 – Compression ratio

$D = 100$ mm, $L = 120$ mm, $V_c = 0.1$ L. $V_s = (\pi/4)(0.1)^2(0.12) = 9.42\times10^{-4}$ m^3 = 0.942 L. $r = (0.942 + 0.1)/0.1 = 10.4$.

❖Problem 2 – Brake power

Torque 150 N·m at 3000 rpm: $BP = 2\pi \times 3000 \times 150 / 60{,}000 = 47.1$ kW.

❖Problem 3 – Efficiencies

Fuel 6 kg/h, CV = 42,000 kJ/kg → heat input = 6×42000/3600 = 70 kW; BP = 21 kW → $\eta_{bth} = 30$ %. If IP = 25 kW → $\eta_{mech} = 21/25 = 84$ %.

❖Problem 4 – Refrigerator

$Q_L = 3.5$ kW, $W = 1$ kW → COP = 3.5. Cooling capacity = 1 TR.

❖Problem 5 – Carnot COP

$T_L = 263$ K, $T_H = 303$ K → COP = 263/40 = 6.6 (practical ≈ 3).

20. Important Questions

- Principles of casting, forming, joining (welding) & machining. Write about CNC, 3D printing, smart manufacturing.
- Working of fire-tube/ water-tube boilers; boiler mountings & accessories.
- Draw P–V diagrams and derive efficiency of Otto and Diesel cycles.
- Explain 4-stroke and 2-stroke engines; compare SI and CI engines.
- VCR cycle with a neat sketch; working of air-conditioning.
- Components of EV and hybrid vehicles.

Learn comparison tables – 2-stroke vs 4-stroke, SI vs CI, fire vs water tube – very frequently asked